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Week 03: Wednesday, AST 5011: Astrophysical Systems

Radiative and Conductive Heat Transfer

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With contributions totally ripped off from Carl Fields (UA), Mike Zingale (SUNY), Cole Miller (UMD), and Abi Nolan (Purdue).

Radiative Transfer

In discussing blackbody radiation and equations of state we assumed LTE as a very good approximation. In deep stellar interiors, LTE implies that the radiation field is nearly isotropic. An exactly isotropic radiation field, however, cannot transport radiant energy.

Anisotropy in the radiation field is therefore required for radiative energy transport.

 specific intensity

Source: HKT Fig. 4.1 — The geometry associated with the specific intensity $I(\theta)$. The position coordinate may either be radius r in a spherically symmetric star or vertical distance z in a plane-parallel “star.” In the latter case, symmetry in the transverse x - and y -coordinates is assumed. The properties of the stellar medium are then independent of azimuthal angle ϕ for either choice of geometry.

This allows us to write the total radiative flux in the z -direction as

$$\mathcal{F} = \int_{4\pi} I(\theta) \cos \theta d\Omega = 2\pi \int_{-1}^{+1} I(\mu) \mu d\mu \quad (\text{erg s}^{-1} \text{ cm}^{-2})$$

where $\mu \equiv \cos \theta$.

If I is constant, the total flux is zero because the same amount of radiation enters as leaves. Radiant energy transport therefore requires that $I(\theta)$ vary with angle; that is, the radiation field must be anisotropic.

We can describe how a beam of radiation changes as it propagates through matter. Radiation may be added to a beam through emission and scattering, described by the mass emission coefficient $j(\theta)$,

$$dI(\text{put into } \theta) = j(\theta) \rho ds$$

where ds is a distance measured along the ray.

Radiation may also be removed from the beam via absorption and scattering,

$$dI(\text{taken out of } \theta) = -\kappa \rho I(\theta) ds$$

Combining these effects gives the net change in specific intensity per unit path length,

$$\frac{1}{\rho} \frac{dI(\theta)}{ds} = j - \kappa I(\theta)$$

This is known as the equation of radiative transfer.

Some definitions:

Source function

$$S_\nu \equiv \frac{j_\nu}{\kappa_\nu}$$

Optical depth

$$\tau_\nu(z) = \tau_{\nu,0} - \int_{z_0}^z \kappa_\nu \rho dz$$

where z_0 is a reference position and $\tau_{\nu,0}$ is the optical depth at that location. For example, if z_0 corresponds to the true surface of the star where $\rho, P \rightarrow 0$, then $\tau_{\nu,0} = 0$. The minus sign ensures that optical depth increases inward.

The Diffusion Equation

In optically thick regions where the photon mean free path is much smaller than the local scale height, the radiation field is nearly isotropic and radiative transfer may be treated as a diffusion process.

To derive the diffusion approximation, we introduce the Rosseland mean opacity κ , which allows us to remove the frequency dependence. The radiative flux may then be written as

$$\mathcal{F}(r) = -\frac{4ac}{3} \frac{1}{\kappa \rho} T^3 \frac{dT}{dr} \equiv -\frac{c}{3\kappa \rho} \frac{d(aT^4)}{dr}$$

The corresponding luminosity is

$$\mathcal{L}(r) = 4\pi r^2 \mathcal{F} = -\frac{16\pi a c r^2}{3\kappa \rho} T^3 \frac{dT}{dr} \equiv -\frac{4\pi a c r^2}{3\kappa \rho} \frac{dT^4}{dr}$$

Using hydrostatic equilibrium,

$$\frac{dP}{dr} = -\frac{GM(r)\rho}{r^2}$$

this can be written in Lagrangian form as

$$\mathcal{L}(r) = -\frac{16\pi acG}{3} \frac{T^4}{P\kappa} M(r) \nabla$$

where we have introduced the logarithmic temperature gradient

$$\nabla \equiv \frac{d \ln T}{d \ln P} = -\frac{r^2 P}{GM(r)\rho} \frac{1}{T} \frac{dT}{dr} \equiv \nabla_{\text{actual}}$$

Finally, we define the radiative temperature gradient,

$$\nabla_{\text{rad}} \equiv \left(\frac{d \ln T}{d \ln P} \right)_{\text{rad}} = \frac{3}{16\pi acG} \frac{P\kappa}{T^4} \frac{L_{\text{tot}}(r)}{M(r)} = \frac{3r^2}{4\pi acG} \frac{P\kappa}{T^4} \frac{\mathcal{F}_{\text{tot}}}{M(r)}$$

A Simple Atmosphere

In a plane-parallel, grey atmosphere in hydrostatic equilibrium, the photosphere is conventionally defined to lie at an optical depth $\tau = 2/3$. The pressure at this location may be written as

$$P(\tau_P) = \frac{2}{3} \frac{g_s}{\kappa_P} \left(1 + \frac{\kappa_P L}{4\pi cGM} \right)$$

The second term represents the correction due to radiation pressure and is typically small except in the most massive and luminous stars. It may be approximated as

$$\frac{\kappa_P L}{4\pi cGM} = 7.8 \times 10^{-5} \kappa_P \left(\frac{L}{L_{\odot}} \right) \left(\frac{M}{M_{\odot}} \right)^{-1}$$

We can also estimate the luminosity at which radiative forces balance gravity. This defines the Eddington critical luminosity,

$$L_{\text{Edd}} = \frac{4\pi cGM}{\kappa_P}$$

In-Class Exercise: Radiative Diffusion in Stars

In the diffusion approximation, the radiative luminosity is given by:

$$\mathcal{L}(r) = -\frac{16\pi a c r^2}{3\kappa\rho} T^3 \frac{dT}{dr}$$

We can explore how the luminosity changes with radius in a simple stellar envelope.

Part 1: Compute dT/dr

► Solution

Part 2: Compute Radiative Luminosity

► Solution

Part 3: Plot the luminosity profile

► Solution

Part 4: Things to try

- How does increasing the opacity affect the luminosity in the envelope?
- What happens to the luminosity profile if the temperature gradient is steeper (e.g., $T \propto r^{-1}$)?
- Compute the radiative gradient $\nabla_{\text{rad}} = \frac{d \ln T}{d \ln P}$ if pressure is $P = \rho k_B T / \mu m_u$ and compare to the adiabatic gradient $\nabla_{\text{ad}} \approx 0.4$. Where in the envelope might convection occur?

► Solution

Formal Solution to the Radiative Transfer Equation

The differential form of the radiative transfer equation is

$$\frac{dI_\nu}{d\tau_\nu} = I_\nu - S_\nu,$$

where I_ν is the specific intensity at frequency ν , S_ν is the source function, and τ_ν is the optical depth along the ray.

The formal solution of this equation, which gives the intensity emerging from an optical depth τ_ν , is

$$I_\nu(\tau_\nu) = I_\nu(0) e^{-\tau_\nu} + \int_0^{\tau_\nu} S_\nu(\tau') e^{-(\tau_\nu - \tau')} d\tau'.$$

This expression can be interpreted as:

1. The first term, $I_\nu(0)e^{-\tau_\nu}$, represents the attenuated incoming intensity. Radiation entering at the boundary is exponentially diminished by absorption and scattering as it travels inward.
2. The second term, $\int_0^{\tau_\nu} S_\nu(\tau')e^{-(\tau_\nu-\tau')}d\tau'$, accounts for emission along the path. Each layer contributes to the outgoing intensity, but the contribution is reduced by the optical depth between that layer and the observer.

This formal solution is foundational for:

- Understanding emergent spectra from stellar atmospheres.
- Connecting the source function to the observable intensity.
- Deriving approximate results, such as the diffusion approximation in optically thick regions.

For example, in the optically thick limit, $\tau_\nu \gg 1$, the integral term dominates, and the intensity approaches the source function, $I_\nu \approx S_\nu$, consistent with LTE.

In-Class Exercise: Formal Solution of the Radiative Transfer Equation

Goal

Explore how incoming radiation is attenuated and how local emission contributes to the observed intensity using the formal solution:

$$I_\nu(\tau_\nu) = I_\nu(0)e^{-\tau_\nu} + \int_0^{\tau_\nu} S_\nu(\tau')e^{-(\tau_\nu-\tau')}d\tau'.$$

This exercise will help build intuition for optically thin vs thick regimes.

Instructions

1. Assume a simple source function that increases linearly with optical depth:

$$S_\nu(\tau) = S_0 + k\tau$$

with $S_0 = 1$ and $k = 2$ (arbitrary units).

2. Take a boundary intensity $I_\nu(0) = 0.5$.
3. Evaluate the formal solution numerically for optical depths ranging from $\tau = 0$ to $\tau = 5$.
4. Plot:

- the contribution from the attenuated incoming intensity: $I_\nu(0)e^{-\tau}$
- the contribution from local emission: $\int_0^\tau S_\nu(\tau')e^{-(\tau-\tau')}d\tau'$
- the total intensity $I_\nu(\tau)$

5. Observe how the incoming term decreases and the emission term grows, and how they combine to produce the total intensity.

► **Solution**

Applications and Extensions of the Formal Solution

The formal solution of the radiative transfer equation

$$I_\nu(\tau_\nu) = I_\nu(0)e^{-\tau_\nu} + \int_0^{\tau_\nu} S_\nu(\tau')e^{-(\tau_\nu-\tau')}d\tau'$$

is particularly useful for understanding stellar atmospheres:

- It explains limb darkening, because different lines of sight probe different optical depths. Near the edge of the stellar disk, we see layers with smaller τ , where the intensity is less than at the center.
- It provides the basis for connecting local source properties (S_ν) to observable intensity (I_ν).

Mean Intensities

To simplify radiative transfer calculations, we define angular moments of the intensity:

$$J_\nu = \frac{1}{4\pi} \int I_\nu d\Omega \quad (\text{mean intensity})$$

$$H_\nu = \frac{1}{4\pi} \int I_\nu \cos \theta d\Omega \quad (\text{flux per unit solid angle})$$

$$K_\nu = \frac{1}{4\pi} \int I_\nu \cos^2 \theta d\Omega \quad (\text{second moment})$$

The Eddington approximation assumes

$$K_\nu = \frac{J_\nu}{3},$$

which provides a simple closure relation to move from the full transfer equation to the diffusion approximation in optically thick regions.

LTE and non-LTE

- In LTE, the source function is simply the Planck function: $S_\nu = B_\nu(T)$. This is valid when collisions dominate over radiative processes.
- Non-LTE occurs when the radiation field is strongly anisotropic or the optical depth is low, and the populations of atomic levels are no longer described by the local temperature.
- The diffusion approximation relies on large optical depth ($\tau \gg 1$) so that the radiation field is nearly isotropic, justifying $K_\nu \approx J_\nu/3$.

In-Class Exercise: Mean Intensities and the Eddington Approximation

Goal

From the intensity $I_\nu(\tau, \mu)$ along different angles, compute the angular moments:

- Mean intensity: $J_\nu = \frac{1}{2} \int_{-1}^1 I_\nu(\mu) d\mu$
- Flux per unit solid angle: $H_\nu = \frac{1}{2} \int_{-1}^1 \mu I_\nu(\mu) d\mu$
- Second moment: $K_\nu = \frac{1}{2} \int_{-1}^1 \mu^2 I_\nu(\mu) d\mu$

Compare K_ν and J_ν to test the Eddington approximation $K_\nu \approx J_\nu/3$.

Instructions

1. Compute the formal solution for several angles, i.e., $\mu = \cos \theta = 0.2, 0.5, 0.8, 1.0$.
2. For each τ , compute J_ν , H_ν , and K_ν using a discrete sum over μ with equal weights (simple quadrature).
3. Plot K_ν/J_ν vs τ and check if it approaches 1/3 at large τ .

► Solution

Radiative Opacity Sources

Radiative opacity describes the efficiency with which matter interacts with radiation through absorption and scattering. Different physical processes dominate in different temperature and density regimes.

Electron Scattering

Consider a beam of photons incident upon a collection of stationary electrons. Here the photon flux is defined as the number of photons per unit area per unit time. The probability that a photon is scattered out of the beam is described by the scattering cross section σ ,

$$\sigma = \frac{\text{number of scattering events per unit time per target}}{\text{incident photon flux}} \quad (\text{cm}^2)$$

The opacity is related to the cross section and the number density of scatterers. For electron scattering,

$$\kappa = \frac{\sigma n_e}{\rho}$$

For photon energies much smaller than the electron rest mass energy, $kT \ll m_e c^2$ (or $T \ll 5.93 \times 10^9$ K), the scattering cross section is well described by Thomson scattering,

$$\sigma_e = \frac{8\pi}{3} \left(\frac{e^2}{m_e c^2} \right)^2 = 0.6652 \times 10^{-24} \text{ cm}^2$$

Assuming the material is fully ionized, the electron-scattering opacity becomes

$$\kappa_e = 0.2(1 + X) \text{ cm}^2 \text{ g}^{-1}$$

Electron scattering is independent of temperature and density and therefore dominates in hot, ionized stellar interiors.

Free–Free Absorption (Inverse Bremsstrahlung)

For free–free absorption we assume LTE, so that

$$\frac{j}{\kappa} = S = B(T)$$

and that the opacity is due entirely to free–free processes. This allows us to write

$$\kappa = \frac{j}{B(T)} = \frac{\pi j}{\sigma T^4}$$

Substituting the free–free emissivity gives

$$\kappa_{\text{ff}} \approx 4 \times 10^{-24} \frac{Z_c^2 n_e n_I T^{-3.5}}{\rho} \propto \rho T^{-3.5}$$

This expression is only approximately correct. To compute the Rosseland mean opacity properly, the frequency dependence must be integrated explicitly (see HKT Eq. 4.22).

A commonly used simplified form is

$$\kappa_{\text{ff}} \approx 10^{23} \frac{\rho}{\mu_e} \frac{Z_c^2}{\mu_I} T^{-3.5}$$

Free–free absorption requires the presence of free electrons. In a mixture of hydrogen and helium with trace metals, free–free opacity becomes negligible below temperatures of order 10^4 K, unless the density is sufficiently high (see the hydrogen half-ionization curve in Fig. 3.10).

The strongest dependence in κ_{ff} is on temperature.

Bound–Free and Bound–Bound Absorption

Bound–free absorption occurs when a photon is absorbed by a bound electron and has sufficient energy to ionize the atom or ion.

An approximate expression for the bound–free opacity is

$$\kappa_{\text{bf}} \approx 4 \times 10^{25} Z(1 + X) \rho T^{-3.5}$$

This expression should not be used at temperatures much below $T \approx 10^4$ K, since most photons are not energetic enough to ionize bound electrons and the ionization fraction drops rapidly.

Bound–bound opacity arises from photon-induced transitions between discrete bound states of atoms or ions. While individual lines can be very strong, the Rosseland mean contribution is often smaller than that of free–free or bound–free opacity in stellar interiors.

H[−] Opacity and Other Sources

One of the most important opacity sources in cool stars is absorption due to the negative hydrogen ion, H[−], through both free–free and bound–free transitions.

Because of the large polarizability of neutral hydrogen, it can bind an extra electron with an ionization potential of 0.75 eV. The resulting opacity depends sensitively on both temperature and metal abundance.

An approximate expression for the H[−] opacity is

$$\kappa_{\text{H}^-} \approx 2.5 \times 10^{-31} \left(\frac{Z}{0.02} \right) \rho^{1/2} T^9$$

Unlike Kramers' opacity laws, H[−] opacity increases very steeply with temperature up to about 10^4 K. Above this temperature, Kramers' opacity and electron scattering dominate, and most H[−] ions are destroyed.

In very cool stars with effective temperatures below ~ 3000 K, molecular absorption and dust or grain opacity become important.

Rosseland vs Planck Mean Opacities and Line Blanketing

In stellar interiors and atmospheres, opacities are frequency-dependent, κ_ν , which can vary by many orders of magnitude across lines, edges, and continua. To simplify calculations, we define mean opacities by averaging over the spectrum:

Rosseland Mean Opacity

The Rosseland mean is a harmonic mean weighted by the derivative of the Planck function:

$$\frac{1}{\kappa_R} = \frac{\int_0^\infty \frac{1}{\kappa_\nu} \frac{\partial B_\nu}{\partial T} d\nu}{\int_0^\infty \frac{\partial B_\nu}{\partial T} d\nu}.$$

- Weighted towards frequencies where the medium is most transparent (small κ_ν).
- Appropriate for radiative diffusion in optically thick regions**, where photons undergo many scatterings before escaping.
- Used in stellar interiors to compute the radiative flux: $\mathcal{F} = -\frac{4ac}{3\kappa_R\rho} T^3 \frac{dT}{dr}$.

Planck Mean Opacity

The Planck mean is an arithmetic mean weighted by the Planck function itself:

$$\kappa_P = \frac{\int_0^\infty \kappa_\nu B_\nu d\nu}{\int_0^\infty B_\nu d\nu}.$$

- Weighted towards frequencies where most energy is emitted.
- Appropriate for radiative heating/cooling rates, e.g., in optically thin regions.
- Less useful for diffusion calculations in optically thick layers.

Line Blanketing and Bound–Bound Opacity

- Bound–bound transitions produce sharp spectral lines.
- In the Rosseland mean, these lines are largely averaged out because the harmonic mean emphasizes transparent regions, so narrow absorption lines have little effect on κ_R .
- In contrast, the Planck mean is more sensitive to line absorption since it weights by the emitted flux.
- This effect, called line blanketing, modifies the emergent spectrum and the radiative energy transport in stellar atmospheres, especially in cooler stars or metal-rich compositions.

In-Class Exercise: Radiative Opacity Sources

Goal

The goal of this exercise is to understand which opacity source dominates under different physical conditions by computing and comparing approximate opacity scalings.

You will:

- compute electron scattering, free–free, and H^- opacities,
- explore how they depend on temperature and density,
- identify which opacity dominates in different stellar regimes.

You do *not* need to worry about precise numerical accuracy — focus on trends and relative magnitudes.

Given Approximations

Use the following approximate opacity formulas (cgs units throughout):

Electron scattering:

$$\kappa_e = 0.2(1 + X)$$

Free-free opacity:

$$\kappa_{\text{ff}} \approx 10^{23} \frac{\rho}{\mu_e} \frac{Z_c^2}{\mu_I} T^{-3.5}$$

H⁻ opacity:

$$\kappa_{\text{H}^-} \approx 2.5 \times 10^{-31} \left(\frac{Z}{0.02} \right) \rho^{1/2} T^9$$

Assume:

- $X = 0.70$
- $Z = 0.02$
- $\mu_e = 1.2$
- $\mu_I = 1.3$
- $Z_c = 1$

Part 1: Write Opacity Functions

Write Python functions for:

- `kappa_es(T, rho)`
- `kappa_ff(T, rho)`
- `kappa_hminus(T, rho)`

Each function should return the opacity in $\text{cm}^2 \text{g}^{-1}$.

Part 2: Explore Temperature Dependence

Fix the density at:

$$\rho = 10^{-7} \text{ g cm}^{-3}$$

Evaluate all three opacities for temperatures between:

$$3 \times 10^3 \text{ K} \leq T \leq 3 \times 10^5 \text{ K}$$

Make a log-log plot of opacity versus temperature.

Questions:

1. Which opacity dominates at low temperature?
2. Which dominates at high temperature?
3. Over what temperature range is H^- important?

Part 3: Explore Density Dependence

Fix the temperature at:

$$T = 10^4 \text{ K}$$

Evaluate the opacities for densities between:

$$10^{-10} \text{ g cm}^{-3} \leq \rho \leq 10^{-4} \text{ g cm}^{-3}$$

Make a log–log plot of opacity versus density.

Questions:

1. Which opacity depends most strongly on density?
2. Which opacity is independent of density?
3. How does this relate to where these processes dominate in stars?

Part 4: Physical Interpretation (Conceptual)

Answer briefly:

1. Why does electron scattering dominate in massive, hot stars?
2. Why is H^- opacity confined to a narrow temperature range?
3. Why does the Rosseland mean tend to suppress the importance of sharp spectral lines?

► Solution

In-Class Exercise: Stellar Opacities

In this exercise, we will explore the different contributions to the opacity in stars and see how they depend on density and temperature.

Part 1: Conceptual Questions

1. Which opacity source dominates in:
 - a hot, massive star ($T > 10^6 \text{ K}$, fully ionized)?
 - a solar-like star ($T \approx 10^6 \text{ K}$, partially ionized H and He)?
 - a cool star ($T < 5000 \text{ K}$, low mass)?
2. Explain why electron scattering opacity is independent of temperature for fully ionized gas, but free-free opacity strongly depends on T.

3. For cool stars with $T < 10^4$ K, why does H^- opacity become important? What happens above 10^4 K?

Part 2: A simplified stellar envelope

We will compute and compare opacities for a simplified stellar envelope.

► Solution

Step 3: Analysis

- Identify the temperature ranges where each opacity dominates.
- Describe qualitatively how the total opacity varies with T and why.

► Solution

Heat Transfer by Conduction

In addition to radiative transfer, energy may also be transported by conduction. Beginning from Fick's law of diffusion,

$$\mathcal{F} = -\mathcal{D}_e \frac{dT}{dr}$$

we can rewrite this expression in a form analogous to radiative diffusion by defining a conductive opacity,

$$\kappa_{\text{cond}} = \frac{4acT^3}{3\mathcal{D}_e\rho}$$

The conductive flux can then be written as

$$\mathcal{F}_{\text{cond}} = -\frac{4ac}{3\kappa_{\text{cond}}\rho} T^3 \frac{dT}{dr}$$

An approximate expression for the conductive opacity is

$$\kappa_{\text{cond}} \approx 4 \times 10^{-8} \frac{\mu_e^2}{\mu_I} Z_c^2 \left(\frac{T}{\rho} \right)^2$$

When multiple transport mechanisms are present, the effective opacity is determined by the smaller of the radiative and conductive opacities. In normal stars, κ_{cond} is very large and conduction is negligible. In dense, degenerate matter such as white dwarfs, conduction dominates energy transport.

Electron Conduction in Degenerate Matter

In addition to radiative transfer, electron conduction can be an important energy transport mechanism in dense degenerate matter, such as white dwarfs or neutron star crusts. Key points:

Microscopic Origin

- In degenerate matter, electrons form a Fermi sea, filling energy states up to the Fermi energy E_F .
- Pauli blocking prevents electrons below E_F from scattering, leaving only electrons near the Fermi surface to carry heat.
- As a result, thermal conduction is dominated by these degenerate electrons, which can have very high velocities and long mean free paths.

Conductive Opacity

The approximate conductive opacity is:

$$\kappa_{\text{cond}} \approx 4 \times 10^{-8} \frac{\mu_e^2}{\mu_I} Z_c^2 \left(\frac{T}{\rho} \right)^2$$

- In normal stars, κ_{cond} is huge $\rightarrow$ conduction is negligible.
- In white dwarfs or other degenerate matter, κ_{cond} is small $\rightarrow$ conduction dominates.

Temperature Dependence

- Degenerate electrons: conductivity is roughly independent of temperature in the fully degenerate regime. The T^2 dependence in the simplified opacity formula is valid for partially degenerate conditions.
- Non-degenerate electrons: conductivity scales roughly as $T^{5/2}/\rho$ (Spitzer conductivity) because higher temperature increases electron velocities and collision rates.

Comparison to Neutron Star Crusts

- In neutron star crusts, conduction is dominated by ultra-relativistic, degenerate electrons.
- Conductive flux can exceed radiative flux by orders of magnitude.
- Temperature gradients are very shallow, and heat from the core can propagate efficiently to the surface.

In-Class Assignment: Heat Transfer by Conduction

Goal

Explore conductive flux vs radiative flux in stars and degenerate objects. Students will:

1. Compute conductive opacity using the given approximation.
2. Compare it to a simple radiative opacity.
3. Determine under which conditions conduction dominates.

Given Formulas

Conductive opacity approximation:

$$\kappa_{\text{cond}} \approx 4 \times 10^{-8} \frac{\mu_e^2}{\mu_I} Z_c^2 \left(\frac{T}{\rho} \right)^2$$

Radiative opacity (simplified Kramers law for illustration):

$$\kappa_{\text{rad}} \approx 10^{23} \frac{\rho}{\mu_e} \frac{Z_c^2}{\mu_I} T^{-3.5}$$

Conductive flux:

$$\mathcal{F}_{\text{cond}} = -\frac{4ac}{3\kappa_{\text{cond}}\rho} T^3 \frac{dT}{dr}$$

Radiative flux:

$$\mathcal{F}_{\text{rad}} = -\frac{4ac}{3\kappa_{\text{rad}}\rho} T^3 \frac{dT}{dr}$$

Use constants:

- $\mu_e = 1.2, \mu_I = 1.3, Z_c = 1$
- Temperature range: $10^6 - 10^8$ K (typical white dwarf interiors)
- Density range: $10^3 - 10^7$ g cm⁻³

Part 1: Write Opacity Functions

Write Python functions:

- `kappa_cond(T, rho)`
- `kappa_rad(T, rho)`

Both should return κ in cm²/g.

Part 2: Compare Opacities

For the temperature-density ranges above:

1. Compute both opacities.
2. Determine which is smaller (the dominant transport mechanism).
3. Make a log-log plot of κ vs ρ for a fixed $T = 10^7$ K.

4. Make a log-log plot of κ vs T for a fixed $\rho = 10^5 \text{ g cm}^{-3}$.

Part 3: Conceptual Questions

1. At what densities does conduction start to dominate over radiation?
2. How does the relative importance of conduction change with temperature?
3. Why is conduction negligible in normal stars but important in white dwarfs?

► **Solution**

In-Class Assignment: Exploring Conditions at the Photosphere

We saw in HKT 4.3 that we can use the equation of HSE to calculate photospheric pressure as a function of g and τ .

Eq. 4.48

$$P(\tau_P) = \frac{2}{3} \frac{g_S}{\kappa_P} \left(1 + \frac{\kappa_P \mathcal{L}}{4\pi c G \mathcal{M}} \right)$$

where $g_s = GM/\mathcal{R}^2$

and Eq. 4.54

$$P(\tau_P) \approx \frac{2}{3} \frac{g_S}{\kappa_P}$$

The model files are again:

- $1M_\odot$: [M1_default_profile8.data](#); [M1_default_profile218.data](#)
- $15M_\odot$: [M15_aprox21_profile8.data](#); [M15_aprox21_profile19.data](#);

a

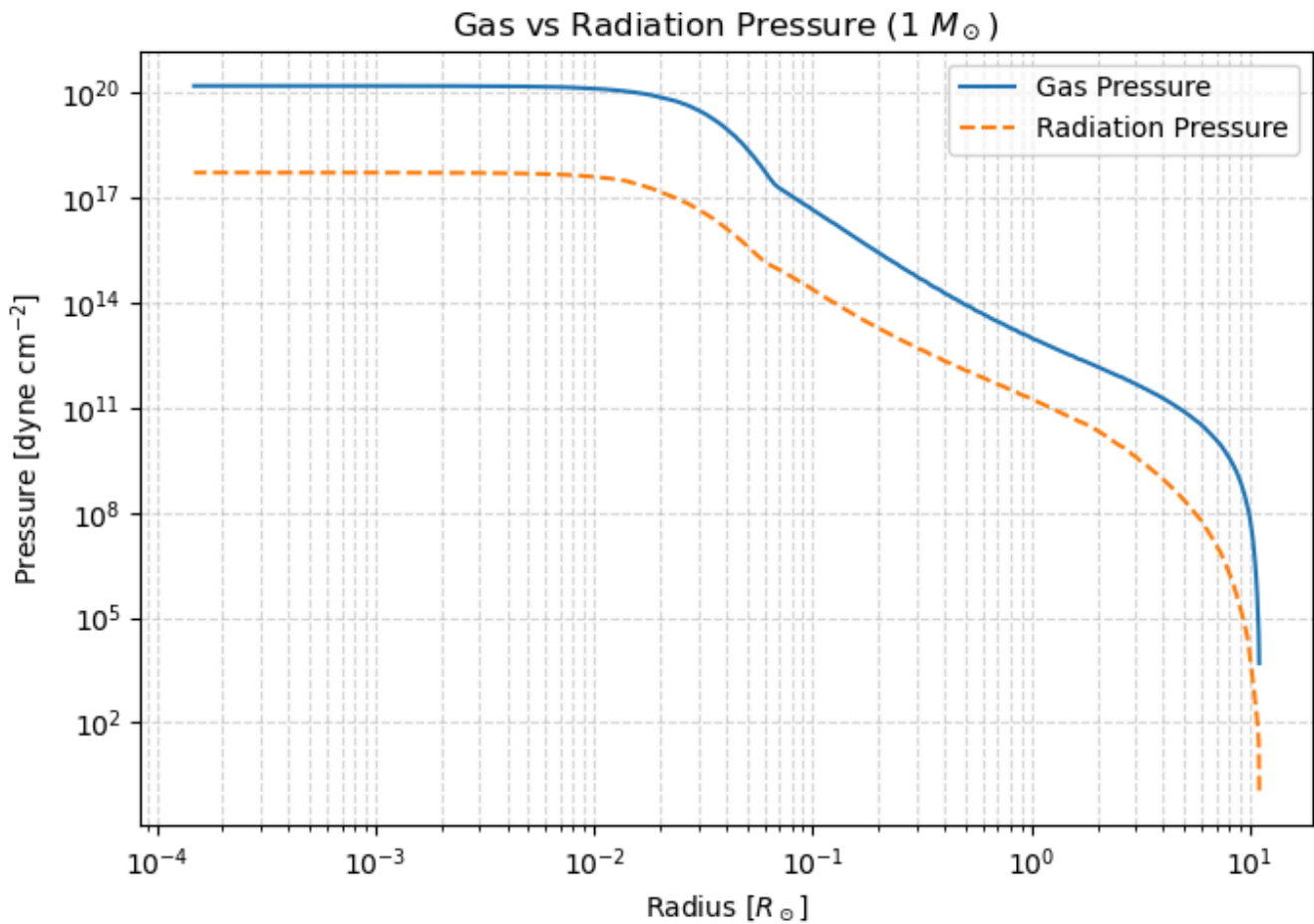
Using your $1 M_\odot$ solar mass model, try to find an evolutionary phase where the radiation pressure term will be largest, though still small compared to the gas pressure. Justify why you've chosen the evolutionary phase that you did.

Plot both the gas pressure and radiation pressure term profiles at this epoch. How good of an approximation is Eq. 4.54?

```

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ntaxWarning: invalid escape sequence '\o'
plt.title("Gas vs Radiation Pressure (1  $M_{\odot}$ )")

```



Maximum Prad / Pgas = $1.783\text{e-}02$
 Median Prad / Pgas = $2.822\text{e-}03$

So even at its largest, radiation pressure contributes only a few percent of the total pressure, and most of the star sits well below the percent level.

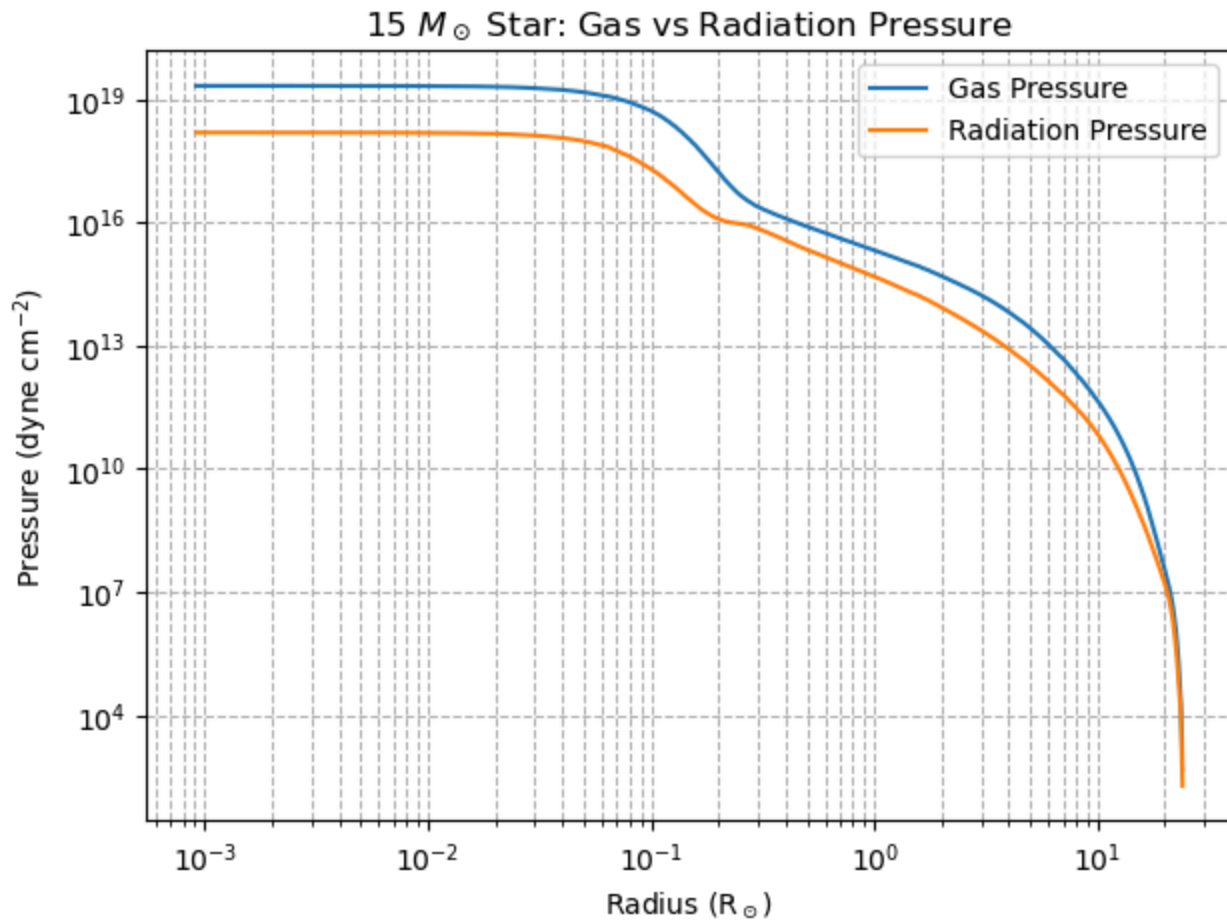
b

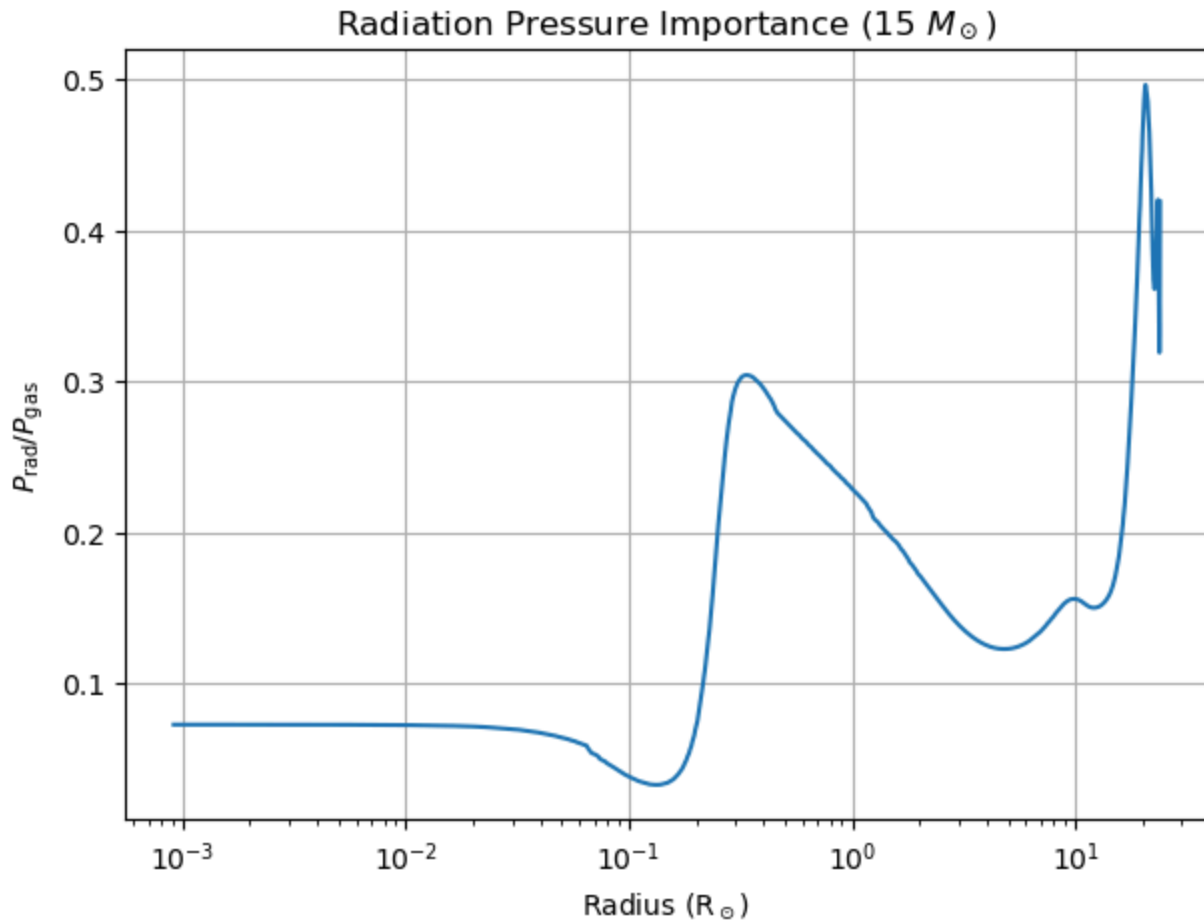
Now return to the $15 M_{\odot}$ models. Find a profile where the radiation pressure term is significant. Plot both the gas pressure and radiation pressure term profiles at this epoch. How good of an approximation is Eq. 4.54? What about 4.48?

```

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taxWarning: invalid escape sequence '\o'
    plt.xlabel("Radius (R$_{odot}$)")
/var/folders/fn/00zl2mgj5w1_bhvyzl0w1tp00000gn/T/ipykernel_47274/269240811.py:57: Syn
taxWarning: invalid escape sequence '\o'
    plt.xlabel("Radius (R$_{odot}$)")
Maximum Prad / Pgas = 4.962e-01
Median Prad / Pgas = 1.738e-01

```





How good is Eq. 4.54?

Not good.

Eq. 4.54 assumes:

$$P \approx P_{\text{gas}}$$

However, when

$$\frac{P_{\text{rad}}}{P_{\text{gas}}} \gtrsim 0.1,$$

this approximation can be wrong at the 10–100% level, especially in the stellar interior.

So for massive stars:

- Eq. 4.54 breaks down
- Neglecting radiation pressure significantly underestimates the total pressure

How good is Eq. 4.48?

Much better — and necessary...

Eq. 4.48 explicitly includes the radiation pressure correction:

$$P(\tau_P) = \frac{2}{3} \frac{g}{\kappa} \left(1 + \frac{\kappa L}{4\pi c G M} \right)$$

For massive stars:

- L/M is large
- The correction term can be order unity
- Radiation pressure provides real support against gravity

So:

- Eq. 4.48 is the correct equation to use
- It naturally connects to the Eddington limit

Summary

In solar-mass stars, radiation pressure is a small correction; in massive stars, it is a dominant component of hydrostatic support, and ignoring it leads to qualitatively wrong physics.

C

Comment qualitatively on the impact of metallicity on the relative importance of the gas pressure vs radiation pressure terms.

- Higher metallicity (Z) increases the opacity of the stellar material because metals provide additional bound-bound, bound-free, and free-free absorption.
- Increased opacity reduces the efficiency of radiative transport, which can lead to steeper temperature gradients and more contribution from radiation pressure in supporting the star.
- For massive stars, higher metallicity enhances the role of radiation pressure because the star must push against more opaque material to maintain energy transport.
- For low-mass stars, metallicity has a smaller effect on radiation pressure because $P_{\text{rad}} \ll P_{\text{gas}}$ anyway, but higher metallicity can still slightly modify the outer envelope structure.
- Takeaway: Metal-rich massive stars are more radiation-pressure dominated compared to metal-poor massive stars, while for solar-type stars the effect is minor.